Harris Corner Detection

I/p image.

0	0	1	4	9
1	0	5	7	11
1	4	9	12	16
3	8	11	14	16
8	10	15	16	20

Kernel.

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \quad \frac{d}{dx}.$$

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \frac{d}{dy}.$$

Step i)

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

I_x.

x	x	x	x	x
x	4	7	6	x
x	8	8	7	x
x	8	6	5	x
x	x	x	x	x

step ii).

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \frac{d}{dy}$$

l_y.

x	x	x	x	x
x	4	8	8	x
x	8	6	7	x
x	6	6	4	x
x	x	x	x	x

Step iii) form mat $|H|$.

from $l_x \Rightarrow 4^2+7^2+6^2+8^2+8^2+7^2$
$+ 8^2+6^2+5^2 = 403.$

from $l_y = 4^2+8^2+8^2+8^2+6^2+7^2+6^2+6^2+4^2$
$= 381.$

from $l_x \cdot l_y \Rightarrow$
$4\times4 + 8\times7\times6*8+8*8+8*6+8*7$
$+8*6+6*6+5*4 = 385.$

$$H = \begin{bmatrix} 403 & 385 \\ 385 & 381 \end{bmatrix}$$

(AB) 22

M Tu W Th F S Su

Step 4:

$$H = \begin{bmatrix} 403 & 385 \\ 385 & 381 \end{bmatrix}$$

trace => sum_diagonal.

$$C = \det(H) - K \, \text{trace}(H)^2$$

$$= 5318 - 0.04 * (784)^2$$

$$= -19268.24$$

if C is large => corner.
if C is negative => edge.
if C is small => flat.